

2.5.24

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September 2025

Question

If $|\mathbf{a}| = 2$, $|\mathbf{b}| = 7$ and $\mathbf{a} \times \mathbf{b} = 3\hat{i} + 2\hat{j} + 6\hat{k}$, find the angle between $\mathbf{a}$ and $\mathbf{b}$

given data

$$|\mathbf{a}| = 2, \quad |\mathbf{b}| = 7, \quad \mathbf{a} \times \mathbf{b} = 3\hat{i} + 2\hat{j} + 6\hat{k} \quad (1)$$

Formula

Let the directions of **a** and **b** be such that

$$\mathbf{a} = (a_1, a_2, a_3)^\top \quad (2)$$

$$\mathbf{b} = (b_1, b_2, b_3)^\top \quad (3)$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \quad (4)$$

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix}^\top \quad (5)$$

$$|\mathbf{a} \times \mathbf{b}| = \sqrt{3^2 + 2^2 + 6^2} = \sqrt{9 + 4 + 36} = \sqrt{49} = 7 \quad (6)$$

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin \theta \quad (7)$$

where the angle between $\mathbf{a}$ and $\mathbf{b}$ is θ

$$7 = 2 \times 7 \sin \theta \Rightarrow 7 = 14 \sin \theta \Rightarrow \sin \theta = \frac{1}{2} \quad (8)$$

Therefore, the angle between $\mathbf{a}$ and $\mathbf{b}$ is $\theta = 30^\circ$

```
import numpy as np
import matplotlib.pyplot as plt

# Define vectors
vec_a = np.array([2, 0, 0])
vec_b = np.array([0, 6, -2])
cross_vec = np.array([3, 2, 6])

fig = plt.figure(figsize=(8, 8))
ax = fig.add_subplot(111, projection='3d')

origin = np.zeros(3)
```

```
# Plot vectors
ax.quiver(*origin, *vec_a, color='r', label=r'$\vec{a}$',
          linewidth=2)
ax.quiver(*origin, *vec_b, color='b', label=r'$\vec{b}$',
          linewidth=2)
ax.quiver(*origin, *cross_vec, color='g', label=r'$\vec{a} \times \vec{b}$', linewidth=2)

# Label vectors
ax.text(*(vec_a * 1.1), r'$\vec{a}$', color='r', fontsize=12)
ax.text(*(vec_b * 1.1), r'$\vec{b}$', color='b', fontsize=12)
ax.text(*(cross_vec * 1.1), r'$\vec{a} \times \vec{b}$', color='g',
        fontsize=12)
```

```
# Axes settings
ax.set_xlim([-1, 8])
ax.set_ylim([-4, 8])
ax.set_zlim([-4, 8])
ax.set_xlabel('X axis')
ax.set_ylabel('Y axis')
ax.set_zlabel('Z axis')
ax.set_title('Vectors  $\vec{a}$ ,  $\vec{b}$ , and  $\vec{a}$   $\times$   $\vec{b}$  with angle = 30')

ax.grid(True)
ax.legend()

plt.show()
```

C Code

```
#include <stdio.h>
#include <math.h>

int main() {
    // Input magnitudes of vectors |a| and |b|
    double mag_a, mag_b;
    // Input cross product components a x b = (cx, cy, cz)
    double cx, cy, cz;

    // Take input
    scanf("%lf %lf", &mag_a, &mag_b);
    scanf("%lf %lf %lf", &cx, &cy, &cz);

    // Calculate magnitude of cross product vector
    double mag_cross = sqrt(cx*cx + cy*cy + cz*cz);

    // Use formula |a x b| = |a||b| sin(theta) to find sin(theta)
    double sin_theta = mag_cross / (mag_a * mag_b);
```

```
import subprocess

# 1. Compile the C program
subprocess.run([gcc, Angle between two vectors.c, -o, Angle
               between two vectors])

# 2. Run the compiled C program
result = subprocess.run([./Angle between two vectors],
                        capture_output=True, text=True)

# 3. Print the output from the C program
print(result.stdout)
```

Vector Plot: $\angle(\vec{a}, \vec{b}) = 30^\circ$

■ $\vec{a}$ ■ $\vec{b}$ ■ $\vec{a} \times \vec{b}$

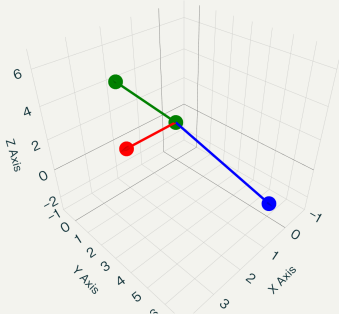


Figure: A vector diagram showing vectors $\vec{a}$, $\vec{b}$, and their cross product with a 30° angle